Capacity-Feasible Clustering for the CVRP on a Neutral-Atom Analog Device

DuQuantum × Pasqal Challenge — Track A — Model write-up

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1 Problem formulation

We state the full Capacitated Vehicle Routing Problem (CVRP) first. Every later modeling choice (Sec. 1.7 onward) is stated as a modification of this model.

1.1 Sets

- $\mathcal{C} = \{1, \ldots, N\}$: customers.
- $\mathcal{V} = \{0\} \cup \mathcal{C}$: all locations; 0 denotes the depot.
- $\mathcal{K} = \{1, \ldots, K\}$: vehicles (identical).

1.2 Parameters

- $p_i = (x_i, y_i) \in [0, 100]^2$, $i \in \mathcal{V}$: location of node i; the depot location p_0 is fixed at $(50, 50)$ before clustering.
  - x_i: x-coordinate of node i.
  - y_i: y-coordinate of node i.
- $d_i \in \mathbb{Z}_{>0}$, $i \in \mathcal{C}$: demand of customer i, with $d_i \leq Q$ (Assumption 5: no customer's demand exceeds the vehicle capacity).
- $Q \in \mathbb{Z}_{>0}$: capacity of every vehicle.
- $K \in \mathbb{Z}_{>0}$: fleet size (fixed; vehicles may be unused).
- $c_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$, $i, j \in \mathcal{V}$: travel cost; symmetric and satisfies the triangle inequality.

1.3 Decision variables

- $a_{ik} \in \{0, 1\}$, $i \in \mathcal{C}$, $k \in \mathcal{K}$: $a_{ik} = 1$ iff vehicle k serves customer i (assignment / clustering).
- $e_{ijk} \in \{0, 1\}$, $i, j \in \mathcal{V}$, $i \neq j$, $k \in \mathcal{K}$: $e_{ijk} = 1$ iff vehicle k travels directly from i to j (edge / routing).

1.4 Objective

Minimize the total distance driven by the fleet:

$$\min_{a,\, e} \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{V}} \sum_{\substack{j \in \mathcal{V} \\ j \neq i}} c_{ij}\, e_{ijk}. \tag{1}$$

1.5 Constraints

$$\sum_{k\in\mathcal{K}} a_{ik} = 1 \qquad \forall i \in \mathcal{C} \qquad \text{(equality: each customer served exactly once)} \qquad \text{(C1)}$$

$$\sum_{i\in\mathcal{C}} d_i\, a_{ik} \le Q \qquad \forall k \in \mathcal{K} \qquad \text{(inequality: capacity)} \qquad \text{(C2)}$$

$$\sum_{j\in\mathcal{V},\, j\neq i} e_{ijk} = a_{ik} \qquad \forall i \in \mathcal{C},\ k \in \mathcal{K} \qquad \text{(equality: leave } i \text{ iff served by } k\text{)} \qquad \text{(C3)}$$

$$\sum_{j\in\mathcal{V},\, j\neq i} e_{jik} = a_{ik} \qquad \forall i \in \mathcal{C},\ k \in \mathcal{K} \qquad \text{(equality: enter } i \text{ iff served by } k\text{)} \qquad \text{(C4)}$$

$$\sum_{j\in\mathcal{C}} e_{0jk} \le 1 \qquad \forall k \in \mathcal{K} \qquad \text{(inequality: at most one trip per vehicle)} \qquad \text{(C5)}$$

$$\sum_{\substack{i,j\in S \\ i\neq j}} e_{ijk} \le |S| - 1 \qquad \forall S \subseteq \mathcal{C},\ |S| \ge 2,\ k \in \mathcal{K} \qquad \text{(inequality: subtour elimination)} \qquad \text{(C6)}$$

$$e_{ijk},\ a_{ik} \in \{0,1\} \qquad \text{(binary domains)} \qquad \text{(C7)}$$

Remarks. (i) Flow balance at the depot, $\sum_j e_{0jk} = \sum_j e_{j0k}$, is implied by (C3)–(C4): every arc has one tail and one head, and customer in- and out-degrees are equal, so the depot's are too. (ii) (C6) forbids any cycle among customers that avoids the depot; combined with (C3)–(C5), each used vehicle drives exactly one closed tour through the depot. (iii) (C6) has exponentially many members; it is used only for the formulation and the classical reference, never on the device. (iv) The two constraints that matter for the device have different types: (C1) is an equality with a single target value, while (C2) is an inequality satisfied by every load in $[0, Q]$. This asymmetry drives the encoding choices in Sec. 2.

1.6 Assumptions and where they are enforced

#	Assumption (challenge brief, Sec. 2)	Enforced by
1	Depot location fixed before clustering	Parameter p_0
2	Symmetric costs, triangle inequality	Parameter c_{ij} (Euclidean)
3	Single vehicle, single visit, no split delivery	(C1), (C3), (C4)
4	One trip per vehicle, no mid-route reload	(C5), (C6)
5	$d_i \le Q$ for all i	Instance generator (Sec. 3)
6	Fixed K; unused vehicles allowed at zero cost	Set $\mathcal{K}$; (C5) is ≤ 1, not $= 1$

1.7 Capacity-feasible clustering

Under cluster-first/route-second, Track A decides only a; the routing variables e and (C3)–(C6) are solved afterwards by an exact classical TSP, used only to evaluate the CVRP gap. This decomposition is a heuristic (brief, Pitfall 2.1).

Register fleet size. Let $D = \sum_i d_i$ and

$$K_{\min} = \min\{K' \in \{1, \ldots, K\} : \text{some } a \text{ with } K' \text{ vehicles satisfies (C1)–(C2)}\},$$

the exact bin-packing number of the demands; $K_{\min} \ge \lceil D/Q \rceil$, and the inequality can be strict (e.g. three demands of 2 with $Q = 3$ need three bins although $\lceil 6/3 \rceil = 2$). $K_{\min}$ is well defined exactly when the instance is feasible, and an assignment using $K_{\min}$ vehicles is also a valid plan for the full fleet, with vehicles $k > K_{\min}$ unused (Assumption 6). We compute $K_{\min}$ exactly per instance (Sec. 3) and model only $K' = K_{\min}$ vehicles, so the clustering model always has a feasible solution and every feasible shot is a valid CVRP plan. The CVRP reference is computed with the full fleet K, not $K_{\min}$.

Objective. We minimize the within-vehicle pairwise distance:

$$\min_{a} \; f(a) = \sum_{k=1}^{K'} \sum_{i<j} c_{ij}\, a_{ik} a_{jk} \quad \text{s.t.} \quad \text{(C1)},\ \text{(C2)},\ a_{ik} \in \{0,1\},\ k \le K'. \tag{2}$$

Its weights are non-negative, so it maps onto positive Rydberg couplings with no linear term (Sec. 2). For any non-negative pairwise objective, moving a customer to an empty vehicle never increases its value, so some minimizer spreads customers over every available vehicle, whereas by the triangle inequality merging two tours (when capacity permits) never increases route cost; sizing the register to $K_{\min}$ largely removes this mismatch and also minimizes the atom count. The model gap (report, Sec. 5) is measured against the optimum of (2) with $K' = K_{\min}$.

2 Encoding and variable accounting

2.1 What the register minimizes

In the analog (Ising) mode the register evolves under

$$\frac{H(t)}{\hbar} = \sum_m \frac{\Omega(t)}{2} \hat{\sigma}_m^x - \delta(t) \sum_m \hat{n}_m + \sum_{m<m'} U_{mm'}\, \hat{n}_m \hat{n}_{m'}, \qquad U_{mm'} = \frac{C_6}{\|\mathbf{r}_m - \mathbf{r}_{m'}\|^6}, \tag{3}$$

where $\mathbf{r}_m$ is the position of atom m. With $\Omega = 0$ at the end of the sweep every operator is diagonal, so a measured bitstring $n \in \{0,1\}^M$ has the classical cost

$$E(n) = -\delta \sum_m n_m + \sum_{m<m'} U_{mm'}\, n_m n_{m'}. \tag{4}$$

This is a QUBO in which the linear coefficient $-\delta$ is set by the (global) laser detuning and is *identical for every atom*, while the couplings are set by the atom positions and are *strictly positive*, decay as r^{-6}, and cannot be made exactly zero; C_6 is fixed. A target QUBO is therefore directly realizable only if all its couplings are ≥ 0 and all its linear coefficients are equal (up to positive scaling and a constant). With AnalogDevice, $U \le C_6/(5\,\mu\text{m})^6 \approx 55.4$ rad/μs and $|\delta| \le 125.66$ rad/μs.

2.2 Textbook formulation: binary slack

Each capacity inequality becomes $L_k + \sum_b w_b s_{kb} = Q$ with $L_k = \sum_i d_i a_{ik}$, slack bits $s_{kb} \in \{0,1\}$, weights $w_b = 2^b$ and $m_Q = \lceil \log_2(Q+1) \rceil$ bits per vehicle, penalized by $B \sum_k \big(L_k + \sum_b w_b s_{kb} - Q\big)^2$ alongside the assignment penalty. Using $b^2 = b$ for binary variables, expanding gives positive couplings $2Bd_id_j$, $2Bd_iw_b$ and $2Bw_bw_{b'}$ (each vehicle's atoms become fully connected) and linear coefficients $B\,d_i(d_i - 2Q)$ and $B\,w_b(w_b - 2Q)$ that differ from atom to atom. It costs Km_Q extra atoms and is not realizable with a global detuning.

2.3 Our formulation

One atom per assignment variable a_{ik}, $k \le K'$, with

$$H(a) = f(a) + \sum_i A_i \Big(1 - \sum_k a_{ik}\Big)^2 + \lambda_2 \sum_k \big(L_k^2 - 2QL_k\big). \tag{5}$$

Capacity penalty and the choice $\lambda_1 = 0$. The capacity term is a slack-free quadratic penalty. We start from the unbalanced penalization of Montañez-Barrera et al. [1], applied to the leftover room $h_k = Q - L_k$ of each vehicle, where $h_k \ge 0$ means vehicle k is within capacity,

$$P(h_k) = -\lambda_1 h_k + \lambda_2 h_k^2, \qquad \lambda_1, \lambda_2 \ge 0. \tag{6}$$

Substituting $h_k = Q - L_k$,

$$\begin{aligned} P(h_k) &= -\lambda_1(Q - L_k) + \lambda_2(Q - L_k)^2 \\ &= -\lambda_1 Q + \lambda_1 L_k + \lambda_2 Q^2 - 2\lambda_2 Q L_k + \lambda_2 L_k^2 \\ &= \lambda_2 L_k^2 + (\lambda_1 - 2\lambda_2 Q) L_k + (\lambda_2 Q^2 - \lambda_1 Q), \end{aligned}$$

and summing over the K' vehicles, where the constant appears once per vehicle,

$$\sum_k P(h_k) = \lambda_2 \sum_k L_k^2 + (\lambda_1 - 2\lambda_2 Q) \sum_k L_k + K'(\lambda_2 Q^2 - \lambda_1 Q). \tag{7}$$

Under (C1) the total load is fixed:

$$\sum_k L_k = \sum_k \sum_i d_i a_{ik} = \sum_i d_i \sum_k a_{ik} = \sum_i d_i = D. \tag{8}$$

Hence, for every assignment satisfying (C1), (7) equals $\lambda_2 \sum_k L_k^2 + (\lambda_1 - 2\lambda_2 Q)D + K'(\lambda_2 Q^2 - \lambda_1 Q)$, and λ_1 enters only through the constant $\lambda_1(D - K'Q)$. This constant is identical for every valid clustering, so λ_1 cannot change their energy ordering. On states violating (C1), λ_1 only adds $\lambda_1 d_i$ to the linear coefficient of each a_{ik}, which the choice of A_i below would absorb. For any λ_1, once the A_i are large enough that every ground state satisfies (C1), λ_1 does not affect the ground state; we set $\lambda_1 = 0$, for which Sec. 2.4 gives such a bound. This removes one hyperparameter without loss of generality.

By the same argument, the term $-2\lambda_2 Q \sum_k L_k$ is also constant on valid clusterings. We keep it because, up to a constant, it makes the capacity term the squared distance of each load from full capacity, $\lambda_2 \sum_k (L_k^2 - 2QL_k) = \lambda_2 \sum_k (L_k - Q)^2 - \lambda_2 K' Q^2$, and the A_i and δ below are derived with it.

On valid clusterings the penalty therefore acts only through $\lambda_2 \sum_k L_k^2$. With $\sum_k L_k = D$ fixed,

$$\sum_k L_k^2 = \frac{D^2}{K'} + \sum_k \left(L_k - \frac{D}{K'}\right)^2, \tag{9}$$

so it penalizes the spread of loads around their mean D/K'. Because $K' = K_{\min} \geq \lceil D/Q \rceil$, the mean satisfies $D/K' \leq Q$: perfectly balanced loads would be feasible. Integer demands and the competing distance objective $f(a)$ prevent exact balance, so loads stay below Q in most but not all instances. Capacity is therefore soft, and we report the feasibility rate. In terms of atoms, $L_k^2 = \sum_i d_i^2 a_{ik} + 2 \sum_{i<j} d_i d_j \, a_{ik} a_{jk}$ contributes positive couplings $2\lambda_2 d_i d_j$ within each vehicle and linear coefficients $\lambda_2 d_i (d_i - 2Q)$.

Expanding with $a^2 = a$, the linear coefficient of a_{ik} is $-A_i + \lambda_2 d_i (d_i - 2Q)$. Choosing

$$A_i = A + \lambda_2 \Big[d_i(d_i - 2Q) - \min_j d_j(d_j - 2Q) \Big] \;\geq A \tag{10}$$

makes every linear coefficient equal to $-\delta$ with $\delta = A + \lambda_2 \max_j d_j(2Q - d_j)$, so a global detuning suffices and no DMM is needed. The resulting coefficients are:

Term	Coefficient
Linear, every a_{ik}	$-\delta$ (identical)
$a_{ik}a_{il}$, same customer, $k \neq l$	$2A_i$
$a_{ik}a_{jk}$, same vehicle, $i \neq j$	$c_{ij} + 2\lambda_2 d_i d_j$
$a_{ik}a_{jl}$, $i \neq j$, $k \neq l$	0
Constant	$\sum_i A_i$

All couplings are non-negative. The target zeros cannot be realized exactly. On assignments satisfying (C1), however, only the difference between the same-vehicle and cross-vehicle couplings of each customer pair affects the energy ordering, so cross-vehicle couplings need not vanish, only be equal (Sec. 4). Fixing a customer to a vehicle would break the vehicle-relabeling symmetry and save K' atoms, but it would make the linear coefficients non-uniform again, so we do not do it.

2.4 Penalty weights

Let $\Gamma_i = \sum_{j \neq i}(c_{ij} + 2\lambda_2 d_i d_j)$.

Claim. *If $\lambda_2 \geq 0$ and $\forall i$ we have*

$$A_i > \max\big(\Gamma_i,\ \lambda_2 d_i(2Q - d_i)\big), \tag{11}$$

then no ground state of (5) *violates* (C1).

Proof. Let a violate (C1) for some customer i, and let $s = \sum_k a_{ik} \neq 1$. We exhibit a state a', differing from a in the single variable a_{ik}, with $H(a') < H(a)$. Changing a_{ik} alters only three terms of (5): the assignment penalty of customer i, the pairs of f that contain a_{ik}, and the capacity term of vehicle k. Write L_k for the load of vehicle k in a.

Case $s = 0$. Choose any vehicle k and set $a_{ik} = 1$. The assignment penalty of customer i drops from A_i to 0. The objective f increases by $\sum_{j \neq i} a_{jk} c_{ij}$. Since customer i is not on vehicle k, $L_k = \sum_{j \neq i} d_j a_{jk}$, and the capacity term changes by

$$\begin{aligned}
&\lambda_2\big[(L_k + d_i)^2 - 2Q(L_k + d_i)\big] - \lambda_2\big[L_k^2 - 2QL_k\big] \\
&\quad = \lambda_2\big[L_k^2 + 2L_k d_i + d_i^2 - 2QL_k - 2Qd_i - L_k^2 + 2QL_k\big] \\
&\quad = 2\lambda_2 d_i L_k + \lambda_2 d_i(d_i - 2Q) \\
&\quad = \sum_{j \neq i} a_{jk}\, 2\lambda_2 d_i d_j + \lambda_2 d_i(d_i - 2Q).
\end{aligned}$$

Because $a_{jk} \in \{0, 1\}$ and $1 \leq d_i \leq Q$, the total change satisfies

$$\Delta H = -A_i + \sum_{j \neq i} a_{jk}\big(c_{ij} + 2\lambda_2 d_i d_j\big) + \lambda_2 d_i(d_i - 2Q) \leq -A_i + \Gamma_i < 0.$$

Case $s \geq 2$. Choose a vehicle k with $a_{ik} = 1$ and set $a_{ik} = 0$. The assignment penalty of customer i changes by $A_i\big[(s-2)^2 - (s-1)^2\big] = -A_i(2s - 3) \leq -A_i$. The objective f changes by $-\sum_{j \neq i} a_{jk} c_{ij} \leq 0$. The capacity term of vehicle k changes by

$$\begin{aligned}
&\lambda_2\big[(L_k - d_i)^2 - 2Q(L_k - d_i)\big] - \lambda_2\big[L_k^2 - 2QL_k\big] \\
&\quad = \lambda_2\big[L_k^2 - 2L_k d_i + d_i^2 - 2QL_k + 2Qd_i - L_k^2 + 2QL_k\big] \\
&\quad = \lambda_2\big[d_i^2 - 2L_k d_i + 2Qd_i\big] \\
&\quad = \lambda_2 d_i\,(d_i - 2L_k + 2Q) \\
&\quad \leq \lambda_2 d_i\,(d_i - 2d_i + 2Q) = \lambda_2 d_i(2Q - d_i),
\end{aligned}$$

where the inequality uses $L_k \geq d_i$ (customer i is on vehicle k) and $\lambda_2 d_i \geq 0$. Hence $\Delta H \leq -A_i + \lambda_2 d_i(2Q - d_i) < 0$.

In both cases a is not a minimizer of H. Since H is defined on the finite set $\{0, 1\}^{NK'}$, a minimizer exists, and by the above it satisfies (C1). □

Condition (11) is sufficient but loose; a large A_i compresses the objective couplings after scaling to the device.

Chosen weights. We use $\lambda_2 = 10$ and $A = \Gamma_{\max}/2$, with $\Gamma_{\max} = \max_i \Gamma_i$. This choice lies below the bound (11) on all 100 tier-S instances tested, so it is validated empirically rather than proven: exhaustive enumeration of all $2^{NK'}$ bitstrings gave a ground state satisfying (C1) on every instance, with outcomes identical to the rigorous choice; with $A = \Gamma_{\max}/4$, (C1) failed for $\lambda_2 \geq 12$. At $\lambda_2 = 10$ the ground state is the constrained optimum on 79 instances, feasible but suboptimal on 19, and over capacity on 2. No $\lambda_2 \in \{6, 8, 10, 12, 15\}$ removes all overloads: the quadratic penalty grows with load imbalance rather than with overload, so a clustering that exceeds Q by one unit costs only $O(\lambda_2 d_{\max})$ more than a balanced one, while larger λ_2 increases the number of suboptimal ground states. Relative to the rigorous bound, $A = \Gamma_{\max}/2$ reduces the median coupling dynamic range from 18.5 to 11.5. Because λ_2 trades distance against demand, it is retuned for each tier. (Script: `step1_qubo_check.py`; seeds 0–99; results in `step1_results.csv`.) On both over-capacity instances ($\rho = 0.93$), the constrained optimum is the first excited energy level of the QUBO, so it is reachable by sampling and selected by the best-feasible-shot rule.

Device scaling. Let $U_{\max}$ be the largest coupling coefficient of (5). Multiplying (5) by $\sigma = 55.4/U_{\max}$ rad/μs places the strongest coupling at the 5 μm minimum spacing; the final detuning is $\sigma\delta$, which must satisfy $\sigma\delta \leq 125.66$ rad/μs. Under (11), $U_{\max} = 2\max_i A_i$, since every same-vehicle coupling satisfies $c_{ij} + 2\lambda_2 d_i d_j \leq \Gamma_i < A_i$; for penalty weights below the bound, $U_{\max}$ must be computed explicitly. This mapping is used by the fitted embedders (report, Sec. 3); the ladder layout of Sec. 4 sets its detuning from a validity window instead.

2.5 Variable accounting

Encoding	Count	S	M	L	Fits in 80?
One-hot assignment, no constraints	NK	8	30	60	yes
+ binary-encoded slack	$NK + K\lceil \log_2(Q+1) \rceil$	14	45	80	L at the cap
Our formulation	$NK_{\min}$	8†	20–30	30–60	yes

†Instances with $K_{\min} = 1$ (4 atoms) are trivial and are excluded by the generator (Sec. 3).

The atom cap is necessary but not sufficient. The emulators bound what can be run in practice (state-vector methods to roughly 20 atoms, emu-mps to roughly 30–60 at moderate entanglement), and the couplings must be realizable in a plane: the one-hot structure requires, for each customer, a strongly coupled K'-clique, and between vehicles, couplings that should vanish. We ran tier S, where $K_{\min} = 2$ on every instance (8 atoms). The tier-M and tier-L counts follow from $K_{\min} \in \{2, 3\}$ and $\{2, 3, 4\}$: under the generator's load-ratio band $\rho \geq 0.45$, total demand always exceeds one vehicle's capacity, and $K_{\min} \leq K$.

3 Instances and reference values

Feasibility certification (Task A2). The brief's assumptions do not guarantee feasibility: $D \leq KQ$ is necessary but not sufficient, and deciding feasibility is a bin-packing problem. For each seeded draw of demands we solve bin packing exactly with the $O(N2^N)$ subset dynamic program ($dp[S]$ = lexicographic minimum of bins used and current-bin load), which returns $K_{\min}$ and an explicit assignment into $K_{\min}$ vehicles. A draw is accepted only if $2 \leq K_{\min} \leq K$: $K_{\min} > K$ is infeasible, and $K_{\min} = 1$ is trivial. The certificate shipped with each instance is this assignment. An independent validator checks every location lies in $[0, 100]^2$, $1 \leq d_i \leq Q$ (Assumption 5), (C1) and (C2) on the certificate, that it uses exactly $K_{\min}$ non-empty vehicles with $K_{\min} \leq K$ (hence feasibility for the full fleet), and minimality of $K_{\min}$: either $K_{\min} = \lceil D/Q \rceil$, which is a lower bound, or an exhaustive backtracking search confirms that $K_{\min} - 1$ vehicles cannot pack the demands. Both algorithms reproduce the brief's counterexample and agree on 2000 random packings.

Generator and seeds. Customer locations are uniform in $[0, 100]^2$ with the depot at $(50, 50)$; demands are uniform integers in the tier's range; a draw is accepted if its load ratio ρ lies in $[0.45, 0.95]$ and $2 \leq K_{\min} \leq K$. We use tier-S seeds 0–99 for the penalty weights (Sec. 2.4) and seeds 0–19 for embedding and emulation. Every instance, with its certificate and reference values, is listed in `instances_tierS.json`.

Reference values. Every reference is a proven optimum, found by brute force: for the clustering model, all K'^N assignments; for the CVRP with the full fleet K, all K^N assignments, with each cluster routed by trying every visiting order from the depot.

CPLEX cross-check (tier S). As required by the brief (Sec. 9.3), we solved the 20 emulated tier-S instances with CPLEX 22.2 (community edition, via `docplex`; time limit `cplex maxtime` = 60 s per solve) and compared with brute force. The clustering model was solved with the products $a_{ik}a_{jk}$ linearized by $y_{ijk} \geq a_{ik} + a_{jk} - 1$, $y_{ijk} \geq 0$; the CVRP with the full fleet was solved in the three-index arc formulation (C1)–(C5), with Miller–Tucker–Zemlin load constraints in place of (C6). Every solve ended with status "integer optimal solution", and CPLEX matched brute force on 20 of 20 instances for both problems. (Script: `cplex_check.py`; results in `cplex_check.csv`.)

4 Ladder layout: derivation

Intuition. Coefficient matching reproduced the wanted couplings well, but the cross-vehicle couplings, which should be zero, remained large enough to reorder the low-energy states (report, Sec. 3). Two observations led to the ladder. First, on any assignment satisfying (C1), each customer has exactly one excited atom, so for each customer pair exactly one of its four couplings is active. Shifting every cross-vehicle coupling of a pair by the same constant therefore changes all valid energies equally: cross-vehicle couplings need to be *equal*, not zero. Second, equality can be forced by symmetry. If every customer's two atoms are mirror images across one common line, the two cross-vehicle distances of any pair coincide, and so do the two same-vehicle distances. Placing vehicle 1's atoms on one rail and vehicle 2's on a parallel rail, with a customer's two atoms at the same position along the rails, achieves this.

Layout. Vehicle k's atoms lie on the rail $y = (k - \frac{1}{2})b$ with $b = 5$ μm, and customer i's two atoms at position ξ_i along both rails. For every customer pair, the same-vehicle, cross-vehicle and one-hot couplings are, exactly,

$$u_{ij} = \frac{C_6}{|\xi_i - \xi_j|^6}, \qquad v_{ij} = \frac{C_6}{\big(|\xi_i - \xi_j|^2 + b^2\big)^3}, \qquad U_{\mathrm{oh}} = \frac{C_6}{b^6}.$$

Why the ladder works. In a valid clustering each customer has exactly one excited atom, so for each customer pair only one coupling is active: u_{ij} if the two share a vehicle (same rail), v_{ij} if they do not (diagonal). The register energy is then

$$E = -\delta_r N + \sum_{i<j} v_{ij} + \sum_{i<j,\ \text{same vehicle}} \big(u_{ij} - v_{ij}\big).$$

The first two terms are the same for every valid clustering, so the register compares clusterings only through the extra cost $u_{ij} - v_{ij}$ that each same-vehicle pair pays. Our model compares them in the same way: on valid clusterings, (5) is a constant plus $\sum_{\text{same vehicle}} w_{ij}$, with $w_{ij} = c_{ij} + 2\lambda_2 d_i d_j$. So if every pair's extra cost is proportional to its weight, $u_{ij} - v_{ij} = t\, w_{ij}$ for one scale $t > 0$, the register ranks valid clusterings exactly as the model does.

Fitting the layout. The only free choice is where each customer sits along the rails. We choose the positions ξ_i that make $u_{ij} - v_{ij}$ as close as possible to proportional to w_{ij}, minimizing the relative squared mismatch with SciPy's derivative-free Nelder–Mead search from 40 random starts, while keeping atoms at least 5 μm apart, inside the register radius, and inside the validity window of the claim below. The fit uses only couplings, never a list of states. We measure how well it worked by the relative error $\varepsilon_L = \|(u - v)/t - w\|/\|w\|$: 0 would reproduce the model exactly, and our tier-S median is 0.55, because positions on a single line cannot satisfy every pairwise condition at once.

Claim. *Let* $B_i = \sum_{j \neq i} u_{ij}$. *If* $\max_i B_i < \delta_r < U_{\mathrm{oh}}$, *every ground state of the register energy* (4) *satisfies* (C1).

Proof. If both atoms of a customer are excited, de-exciting one of them changes E by $\delta_r - U_{\mathrm{oh}}$ minus its (non-negative) couplings to the other excited atoms, which is at most $\delta_r - U_{\mathrm{oh}} < 0$. Hence a minimizer has no customer with two excited atoms. In such a minimizer, suppose customer i has no excited atom. Every other customer j then has at most one excited atom, whose coupling to either atom of i is u_{ij} or $v_{ij} \leq u_{ij}$. Exciting one of i's atoms therefore changes E by at most $-\delta_r + B_i < 0$, a contradiction. So every customer has exactly one excited atom. □

We fit with $\max_i B_i \leq 0.8\, U_{\mathrm{oh}}$ and set δ_r to the midpoint of the window. Unlike Sec. 2.4, this guarantee holds for the realized register itself, including every residual coupling. Capacity (C2) remains soft. The layout is defined only for $K' = 2$: with three rails in a row, the outer one-hot coupling is $2^6 = 64$ times weaker and the window closes, and with a triangle of atoms per customer the cross-vehicle distances are no longer equal. Its one-dimensional coordinates also cannot represent arbitrary pairwise weights.

References

[1] J. A. Montañez-Barrera, D. Willsch, A. Maldonado-Romo, and K. Michielsen, "Unbalanced penalization: a new approach to encode inequality constraints of combinatorial problems for quantum optimization algorithms," *Quantum Science and Technology* **9**, 025022 (2024). arXiv:2211.13914.